Power BI Lab Report

Cyber Attack Dashboard Analysis

Business Analytics / Descriptive Analytics Course | Power BI Lab Manual

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Practical:	Designing a three-page Cyber Attack dashboard in Power BI (Overview, Global Threats, Incident Analysis)
Dataset:	Global Cybersecurity Threats, 2015 - 2024 (3,000 incident records)
Date:	12 October 2026

Dataset Used: Global Cybersecurity Threats (2015 - 2024)

The dashboard is built on one incident table with **3,000 rows**, one row per recorded cyber incident, covering ten years (2015 - 2024) and many countries. Each row records the attack type, who carried it out, which industry was hit, the weakness that was exploited and the impact (money lost, users affected, hours to resolve).

Columns used in the report

Column	Meaning	Data type
Country	Country where the incident occurred	Text
Year	Year of the incident (2015 - 2024)	Whole number
Attack Type	Phishing, DDoS, Man-in-the-Middle, SQL Injection, Malware, Ransomware	Text
Attack Source	Hacker Group, Insider, Nation-state, Unknown	Text
Target Industry	Healthcare, IT, Government, Banking, Education, Retail, Telecommunications	Text
Security Vulnerability Type	Zero-day, Social Engineering, Unpatched Software, Weak Passwords	Text
Financial Loss (in Million $)	Money lost in the incident	Fixed decimal number
Number of Affected Users	People affected by the incident	Whole number
Incident Resolution Time (in Hours)	Hours taken to resolve the incident	Whole number

Incident table (first 6 rows, as shown on the Incident Analysis page)

Country	Attack Type	Attack Source	Target Industry	Vulnerability	Loss ($M)	Affected Users
Australia	DDoS	Hacker Group	Banking	Weak Passwords	92.66	1,759,968
Australia	DDoS	Hacker Group	Education	Weak Passwords	31.74	724,207
Australia	DDoS	Hacker Group	Government	Zero-day	165.09	1,551,629
Australia	DDoS	Hacker Group	Healthcare	Weak Passwords	77.04	8,517
Australia	DDoS	Hacker Group	Telecommunications	Social Engineering	61.56	767,113
Australia	DDoS	Insider	Banking	Weak Passwords	267.14	1,886,361

Whole dataset totals (Incident Analysis page, no filters): financial loss **$151,478.91 million** (about $151.48 billion), affected users **1,514,052,409**, incidents **3,000**.

Dashboard Design: Pages, Slicers and Measures

Objective: Turn the incident table into an interactive, dark-themed security dashboard that lets a reader filter by year, country and attack type and see where the money, users and risk are concentrated.

Operations performed in Power BI Desktop

1. Imported the incident data with **Home > Get data**, checked the data types in Power Query (text for categories, numbers for loss, users and hours) and clicked **Close & Apply**.

2. Created the KPI measures (Total Incident, Total Finance Loss, Affected Users, Avg Resolution Time) with **Home > New measure**.

3. Added a dark background image, a neon hexagon logo and a left side panel with page buttons: **Overview, Cyber, Global Threats, Attack Analysis, Security, Incident Details**.

4. Added **Year, Country and Attack Type** slicers (dropdown style) to each page so every visual responds to the same choices.

5. Built the visuals for each page (listed below), set titles and data labels, and tested cross-filtering by changing a slicer and clicking chart elements.

Pages and visuals

#	Page	Visuals
1	Cyber Attack Overview	4 KPI cards, line chart (Attack by Year), donut (Attack Type Distribution), column chart (Financial Loss by Attack Type), treemap (Incident by Target Industry), text box (Cyber Risk Briefing)
2	Global Threat Intelligence	Gauge (Incident Resolution Time), arc map (Affected Users by Country), donut (Vulnerability Distribution), funnel (Attack Type Ranked), ribbon chart (Attack Type Ranking by Year), bar chart (Financial Loss by Country)
3	Incident Analysis	Detail table with totals, bar chart (Attack Source Distribution)

Measures used

```
Total Incident = COUNTROWS(Cyber)
Total Finance Loss ($B) = DIVIDE(SUM(Cyber[Financial Loss (in Million $)]), 1000)
Affected Users (B) = DIVIDE(SUM(Cyber[Number of Affected Users]), 1000000000)
Avg Resolution Time (Hrs) = AVERAGE(Cyber[Incident Resolution Time (in Hours)])
```

Outcome: Three linked report pages share the same slicers and a common navigation panel, so a change in Year, Country or Attack Type updates every visual on the page.

Page 1: Cyber Attack Overview

Objective: Give a one-screen summary of incidents, losses, affected users and resolution time, with the attack mix and the industries hit. The Year slicer is set to **2017** in this view.



Figure 1: Cyber Attack Overview page (Year = 2017, Country = All, Attack Type = All).

KPI cards (Year = 2017)

Measure	Value
Total Incident	319
Total Finance Loss ($B)	16.26
Affected Users (B)	0.16
Avg Resolution Time (Hrs)	35.39

Page 1: Overview Breakdown (Year = 2017)

Attack type distribution and financial loss

Attack Type	Share of incidents	Incidents (of 319)	Financial loss ($M)
Phishing	20.06%	64	3,320
DDoS	18.18%	58	3,078
Man-in-the-Middle	18.18%	58	3,086
SQL Injection	15.36%	49	2,256
Malware	14.42%	46	2,141
Ransomware	13.79%	44	2,380
Total	100.00%	319	16,261

Incident counts are the donut percentages applied to 319 incidents. Financial loss values are the column labels on the chart (3.32K = $3,320 million); their total, $16,261 million, matches the $16.26 billion KPI card.

Target industries

The treemap shows Healthcare as the largest block, followed by IT, Government and Banking, then Education, Retail and Telecommunications as the smallest blocks. The Cyber Risk Briefing text states that DDoS has the highest total financial loss, IT has the highest incident exposure and Nation-State attacks have the highest total financial loss by source.

Outcome: The four KPI cards, the donut centre (319) and the attack-type table agree with each other. Phishing is the most frequent attack type and the largest loss in the 2017 view; the other five types are within a few percentage points of each other.

Page 2: Global Threat Intelligence

Objective: Show where attacks land, which weaknesses are exploited and how attack types rank across years. All slicers are set to **All**, so the figures cover 2015 - 2024.



Figure 2: Global Threat Intelligence page (all years, all countries).

Vulnerability distribution (3,000 incidents)

Vulnerability Type	Incidents	Share
Zero-day	785	26.17%
Social Engineering	747	24.90%
Unpatched Software	738	24.60%
Weak Passwords	730	24.33%
Total	3,000	100.00%

Page 2: Ranking and Resolution Time

Attack type ranked (all years)

Rank	Attack Type	Incidents	Share
1	DDoS	531	17.70%
2	Phishing	529	17.63%
3	SQL Injection	503	16.77%
4	Ransomware	493	16.43%
5	Malware	485	16.17%
6	Man-in-the-Middle	459	15.30%
	Total	3,000	100.00%

Share is each count divided by 3,000 incidents. DDoS and Phishing are separated by only two incidents.

Incident resolution time

The gauge shows an average of **36.48 hours** on a scale from 0 to 72.95 hours, so the average sits at the middle of the range.

Financial loss by country

The bar chart ranks countries in this order, highest first: **UK, Brazil, India, France, Japan, Australia, Russia, Germany, USA** (scrollable list). The bars are close in length, so no single country dominates. The **Attack Type Ranking by Year** ribbon chart shows the order of the six attack types changing from year to year between 2015 and 2024.

Outcome: The vulnerability counts add up to 3,000 and so do the attack type counts. The four vulnerability types are almost evenly split (24% to 26%), which shows that no single weakness explains most incidents.

Page 3: Incident Analysis

Objective: Let the reader drill down to individual incidents and see the totals for any combination of Year, Country and Attack Type.

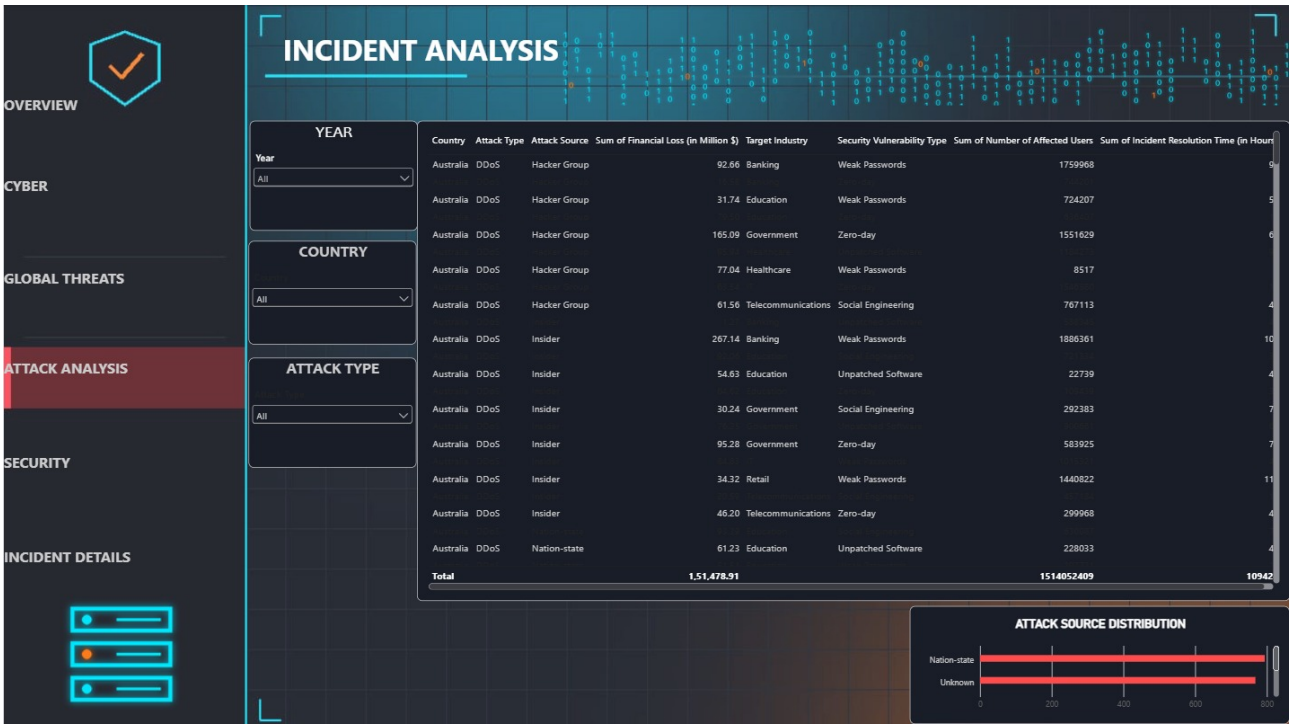


Figure 3: Incident Analysis page (Year = All, Country = All, Attack Type = All).

Table totals (no filters)

Measure	Value
Sum of Financial Loss (in Million $)	151,478.91
Sum of Number of Affected Users	1,514,052,409
Incidents	3,000

The table lists Country, Attack Type, Attack Source, Sum of Financial Loss, Target Industry, Security Vulnerability Type, Sum of Number of Affected Users and Sum of Incident Resolution Time, with a Total row at the bottom. The **Attack Source Distribution** bar chart shows Nation-state and Unknown as the two largest sources, each at roughly 800 incidents (Hacker Group and Insider are further down the scrollable chart).

Outcome: The table total of $151,478.91 million equals the sum of all rows, and the slicers narrow both the table and the source chart together.

Key Findings

Area	Finding
Scale	3,000 incidents caused about $151.48 billion in losses and affected about 1.51 billion users between 2015 and 2024.
Attack types	DDoS (531) and Phishing (529) are the most common; Man-in-the-Middle is the least common (459). The gap between first and last is 72 incidents.
Vulnerabilities	Zero-day is the most exploited weakness (26.17%), but Social Engineering, Unpatched Software and Weak Passwords are each close to 25%.
Industries	Healthcare is the largest block in the 2017 treemap, followed by IT, Government and Banking.
Sources	Nation-state and Unknown sources lead the incident count (about 800 each).
Resolution	Average resolution time is 36.48 hours across all years (35.39 hours in 2017).
Geography	UK, Brazil and India have the highest financial loss, with the other countries close behind.

Conclusion

The dashboard answers the main questions a security team would ask: how many incidents there were, what they cost, which attack types and weaknesses are behind them, where they happen and how long they take to fix. Slicers on every page keep the figures consistent when the view is narrowed to a year, country or attack type.